Dhruv Ramani

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Education

Newton School of Technology (Rishihood University) India
B.Tech in Computer Science and Artificial Intelligence (3rd Year) *Aug 2024 – 2028*

- **Relevant Coursework:** Neural Networks and Deep Learning, Generative AI, Linear Algebra, Statistics, Data Analytics and Visualization

Publications

How Well Does a Kepler-Trained Transit Classifier Transfer to TESS? | *Preprint* Jul 2026

- Sole author. Evaluated cross-mission generalization of a Kepler-trained transit classifier on TESS data, quantifying performance degradation with confidence intervals and assessing threshold adjustments to improve planet recall. https://doi.org/10.5281/zenodo.22132135

Projects

DriftLens – Reward-Hacking Monitoring for RL *(ongoing)* | *PyTorch, SB3, Gymnasium* Sep 2026 – Present

- Instrumented PPO to log per-update telemetry (anchor-gradient alignment, effective rank and subspace angles of parameter updates, behavioral entropy) and built two Gymnasium benchmarks, `ShortcutGrid` and `TippingGrid`, with deliberately exploitable proxy rewards and a separate true-task oracle kept out of the online score.
- Ran multi-seed studies with predeclared gates, frozen thresholds and untouched test layouts, and reported the negative results: the composite score false-alarmed on **3/3 normal controls**, and a proposed benchmark condition failed its AUC ≤ 0.60 gate at **0.731**.
- Built a semantic-twin control: **5/5 paired policies** matched on all 392 learner-visible updates while differing on the hidden objective, showing that proxy-only signals cannot identify reward hacking without trusted outcome audits.

WaterTrace – Satellite Surface-Water Mapping *(ongoing)* | *OpenCV, Rasterio, FastAPI* Oct 2026 – Present

- Built a local pipeline that maps surface water from Sentinel-1 VV/VH radar imagery using dual-band agreement and spatial filtering, keeping uncertain or missing pixels as an explicit unknown state so a missing observation is never read as "no water".
- Evaluated a frozen policy on all 90 Sen1Floods11 test chips (10 flood events, 20.5M labeled pixels): **45.2% IoU** and **75.6% precision** vs. 23.2% and 25.5% for a VH Otsu baseline, at a recall cost (52.9% vs. 72.2%).
- Reported every event with no post-hoc scene selection, including failures: Pakistan and Sri Lanka fell below the baseline's IoU, so pooled precision is not claimed as universal reliability.

Mavis – Real-Time CV Fitness Coach | *MediaPipe, OpenCV, React, TypeScript* Nov 2025 – Present

- Engineered an on-device pose pipeline (33 MediaPipe landmarks → EMA smoothing → stillness-gated calibration → per-exercise state machine) that counts reps and scores each out of 100, with body-relative fault detectors so a stance shift is not misread as a form fault. No video leaves the device.
- Cut the false-fault rate from **61%→5%** (shoulder press) and **46%→10%** (bicep curl) by setting every tolerance from measured distributions over 128 recorded reps; a pytest suite replays landmark fixtures to assert clean reps pass and poor form is still caught.

Exoplanet Detection Pipeline | *Scikit-learn, XGBoost, FastAPI, Next.js* Feb 2025 – Present

- Built an end-to-end transit-detection pipeline over live NASA TESS data (MAST lightcurves, `wotan` detrending, Transit Least Squares) with five physics vetting checks and a plain-English reasoning trail for every verdict.
- Trained a Platt-calibrated RandomForest + XGBoost ensemble on 7,325 Kepler objects: **0.957 ROC-AUC, 0.917 PR-AUC** on 1,465 unseen targets.

Technical Skills

Languages: Python, JavaScript/TypeScript, Dart, SQL
RL / ML: PyTorch, Gymnasium, Scikit-learn, XGBoost, model calibration
Web & Mobile: React.js, FastAPI, Node.js, Tailwind CSS
Libraries & Tools: NumPy, Pandas, OpenCV, Rasterio, MediaPipe, Docker, pytest